

Dr. Umer Farooq

PhD (Computer Science)

Professor | Chairperson, Department of Computing | Researcher

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🎓 Google Scholar 🔗 LinkedIn 🆔 ORCID

🌐 Portfolio



EXECUTIVE SUMMARY

Highly accomplished academic professional with a **PhD in Computer Science** and over **10 years of experience** in teaching and mentoring students at the university level. Adept at developing and delivering comprehensive curricula that meet the needs of diverse student populations. Proven ability to engage students through innovative teaching methods and to inspire a passion for learning in the field of computer science. Demonstrates a strong commitment to academic excellence, continuous professional development, and fostering a collaborative learning environment. Skilled in research, curriculum design, and the application of cutting-edge educational technologies.

EDUCATION & QUALIFICATIONS

PhD in Computer Science

2015 – 2021

National College of Business Administration & Economics (NCBA&E), Lahore — School of Computer Sciences

- **Thesis Title:** *Knowledge Heterogeneity Analysis & Network Model (KHAN) Using Artificial General Intelligence.*
- **Supervisor:** Dr. M. Adnan Khan.
- **Research Contribution:** Proposed the **KHAN model** to address challenges in Industry 4.0, Knowledge Management, Cloud Computing, IoT, and Big Data. Emphasizes transforming data from information to knowledge using cognitive science principles, classifying and clustering knowledge, and managing complex data with machine learning. The model is validated through simulations, supporting system evolution and future research directions.
- **Academic Fields of Interest:** Artificial Intelligence, Machine Learning, Cloud Computing.

M.Phil. (Computer Science)

2007 – 2014

National College of Business Administration & Economics (NCBA&E), Lahore — School of Computer Sciences

BSCS (Bachelor of Science in Computer Science)

2002 – 2006

Shah Abdul Latif University, Khairpur – Penta College of Information Technology, North Nazimabad, Karachi

PROFESSIONAL EXPERIENCE

Professor

23 Dec 2026 – Present

Hamdard University, Main Campus, Karachi

- Delivering advanced graduate and postgraduate courses in Computer Science.
- Leading research initiatives, mentoring PhD/MS scholars, and contributing to international publications.

Chairperson, Department of Computing (FEST)

3 June 2025 – Present

Hamdard University, Karachi

- Heading the Department of Computing under the Faculty of Engineering Sciences & Technology (FEST).
- Overseeing academic planning, faculty management, curriculum design, accreditation, and departmental research strategy.

Associate Professor

26 Dec 2024 – 22 Dec 2026

Hamdard University, Main Campus, Karachi

- Taught advanced computer science courses and supervised MS/PhD research.
- Contributed to departmental governance, curriculum development, and quality enhancement.

Assistant Professor

21 Aug 2024 – Dec 2024

Hamdard University, Main Campus, Karachi

- Taught core and elective courses in Computer Science and Artificial Intelligence.

Assistant Professor

Apr 2021 – Aug 2024

Lahore Garrison University (LGU), Lahore

- Taught graduate and undergraduate courses; supervised MS theses in AI, Deep Learning and NLP.
- Served as **Associate Editor**, LGU Research Journal of Computer Science and IT (LGURJCSIT).
- Coordinated national and international conferences and ORIC research activities.

Lecturer

Nov 2014 – Apr 2021

Lahore Garrison University (LGU), Lahore

- Delivered undergraduate courses across Computer Science programs.
- Contributed to journal editing, conference organization, and research mentorship.

Editor — LGURJCSIT (LGU Research Journal of CS & IT)

Mar 2017 – Aug 2024

Lahore Garrison University, Lahore

Lecturer (System Analyst)

2011 – Jun 2014

NCBA & E, Lahore

- Taught Computer Science courses and supported departmental systems analysis activities.

KEY RESPONSIBILITIES & ESTEEM INDICATORS (2014 – Present)

Leadership & Administration

- **Chairperson**, Department of Computing, FEST, Hamdard University, Karachi.
- **Incharge Program BSCS** (2024 – June 2025), Department of Computing, Hamdard University, Main Campus Karachi.
- **Associate Editor**, Lahore Garrison University Research Journal of Computer Science and IT (LGURJCSIT) — 2017 to 2024.
- **National / International Conference Co-Ordinator** (2017 – 2024).
- **ORIC Team Co-Ordinator**.
- **BIC LGU Member** (2021 – 2022).

Teaching Portfolio

- Advanced Theory of Automata
- Artificial Intelligence
- Artificial Neural Networks
- Database Management Systems
- Machine Learning
- Fuzzy Systems
- Social Network Analysis

Thesis Supervision (MS / Research)

- Real-Time Road Lane Detection and Classification using Deep Learning.
- Road Sign Detection and Classification using Deep Learning.
- Sentiment Analysis: A Semantic Approach to Opinion Mining Using the BERT Method.

RESEARCH AREAS & TECHNICAL EXPERTISE

Core Research	Artificial Intelligence	Machine Learning	Deep Learning	Cloud Computing	IoT
Applied AI	NLP / BERT	CNN	LSTM	Explainable AI (XAI)	Fuzzy Inference
Security	Intrusion Detection	Cyber Security	DDoS Mitigation	Cloud Security	
Knowledge Mgmt.	KHAN Model	Knowledge Heterogeneity	Big Data	Industry 4.0	
Specialised	Smart Traffic Systems	Autonomous Vehicles	Fog-Cloud Computing	Block Chain	
Academic	Curriculum Design	Journal Editing	Conference Co-ordination	Thesis Supervision	

JOURNAL PUBLICATIONS

1. **Autonomous Vehicles Congestion Model: A Transparent LSTM-Based Prediction Model Cooperated with Explainable Artificial Intelligence (EAI).** *Egyptian Informatics Journal*, 29 November 2024. [Category W] [\[Link\]](#)
2. **Enhancing Traffic Flow and Congestion Management in Smart Cities Utilizing SVM-Based Linear Regression Approach.** *International Journal of Advanced and Applied Sciences*, 6 October 2024. [Category X] [\[Link\]](#)
3. **Enhanced and Optimized Industrial Processes Management Using Machine Learning (ML) and Block Chain.** *Pakistan Journal of Scientific Research*, 29 May 2024. [Category Y] [\[Link\]](#)
4. **Floating Non-Parametric Control Charts for Process Parameter Using Computation.** *Journal of Computing & Biomedical Informatics*, 1 June 2024. [Category Y] [\[Link\]](#)
5. **Intellectual Gesticulation Identification Assembly.** *Journal of Computing & Biomedical Informatics*, 1 June 2024. [Category Y] [\[Link\]](#)
6. **Securing Cloud Environments: A Convolutional Neural Network (CNN) Approach to Intrusion Detection System.** *Journal of Computing & Biomedical Informatics*, 1 March 2024. [Category Y] [\[Link\]](#)
7. **A Critical Review of Artificial Intelligence Based Approaches in Intrusion Detection: A Comprehensive Analysis.** *Journal of Engineering*, 15 April 2024. [Category X] [\[Link\]](#)
8. **Intelligent Model for Brain Tumor Identification Using Deep Learning.** *Applied Computational Intelligence and Soft Computing*, 21 January 2022. [Category X] [\[Link\]](#)
9. **An Intelligent Proposed Model for Task Offloading in Fog-Cloud Collaboration Using Logistics Regression.** *Computational Intelligence and Neuroscience*, 25 January 2022. [Category Y] [\[Link\]](#)
10. **Cloud-IoT Integration: Cloud Service Framework for M2M Communication.** *Intelligent Automation & Soft Computing*, 03 September 2021. [Category X] [\[Link\]](#)

11. **Comparative Analysis: Intrusion Detection in Multi-Cloud Environment to Identify Way Forward.** *International Journal of Advanced Trends in Computer Science and Engineering*, June 2021. [Category Y] [\[Link\]](#)
12. **Semantic Analysis of Urdu English Tweets Empowered by Machine Learning.** *Intelligent Automation and Soft Computing*, 26 July 2021. [Category X] [\[Link\]](#)
13. **Heterogenetic Knowledge Classification Using Fuzzy Inference for Unified Data Clusters.** *EAI Endorsed Transactions on Scalable Information Systems*, 13 November 2019. [Category X] [\[Link\]](#)
14. **Detection and Mitigation of DDoS Attack in Cloud Computing Using Machine Learning Algorithm.** *ISeCure*, 2019. [Category X] [\[Link\]](#)
15. **An Adaptive Approach: Smart Traffic Congestion Control System.** *Journal of King Saud University — Computer and Information Sciences*, 23 October 2018. [Category W] [\[Link\]](#)

CONFERENCE PUBLICATIONS

1. N. A. Al-Dmour, **U. Farooq**, I. Sarwar, M. W. Iqbal, M. Aqeel, W. A. Khan, H. A. Hamadi. **Cyber Security Threats on Multifunctional Devices and Mitigation Techniques.** *IEEE 2023 International Conference on Business Analytics for Technology and Security (ICBATS)*, 2023. [\[DOI\]](#)
2. S. E. A. Alnawayseh, T. A. Khan, **U. Farooq**, S. Zulfiqar, S. Khan, A. H. Al-Kassem. **Research Challenges and Future Facet of Cellular Computing.** *IEEE 2023 International Conference on Business Analytics for Technology and Security (ICBATS)*, 2023. [\[DOI\]](#)
3. A. Abualkishik, M. Saleem, **U. Farooq**, M. Asif, M. Hassan, J. A. Malik. **Genetic Algorithm-Based Adaptive FSO Communication Link.** *IEEE 2023 International Conference on Business Analytics for Technology and Security (ICBATS)*, 2023. [\[DOI\]](#)
4. M. S. Munir, K. Parveen, **U. Farooq**, K. Shaalan, A. Z. Abualkishik, A. S. Mohammed. **Use of Different Machine Learning Algorithms for Hate Speech Detection.** *2022 International Conference on Cyber Resilience (ICCR)*, pp. 1–7, 2022. [\[DOI\]](#)
5. M. Mehmood, N. Fatima, A. Z. Abualkishik, **U. Farooq**, A. Ibrahim, K. Agha. **An Empirical Study Exploring the Factors of Stress Level among Doctors.** *2022 International Conference on Cyber Resilience (ICCR)*, pp. 1–8, 2022. [\[DOI\]](#)
6. A. Ali, N. A. Somroo, **U. Farooq**, M. Asif, I. Akour, W. Mansoor. **Smartphone Security Hardening: Threats to Organizational Security and Risk Mitigation.** *2022 International Conference on Cyber Resilience (ICCR)*, pp. 1–12, 2022. [\[DOI\]](#)
7. M. Shahid, K. Munir, S. Muneer, M. Jarrah, **U. Farooq**. **Implementation of ML Algorithm for Mung Bean Classification using Smart Phone.** *2022 International Conference on Business Analytics for Technology and Security (ICBATS)*, 2022. [\[DOI\]](#)

REFERENCES

Available on request.